

APPROVED
BSc (Hons) Games Design and Development
 School of Computing, Engineering and Technology

Version	1.0
Course Type	Undergraduate
Award Type	Bachelor of Science with Honours - BSc (Hons)
Awards	
No Course Award Assigned	
Code	0739
SCQF Level	Level 10
QAA Subject Benchmark	Computing (2022)
SCQF Credit Points	480
ECTS Credit Points	240
Valid From	2027 September

HECoS

The maximum number of HECoS Codes allowed is 3.

Code	Title	Alternate Title	Percentage
101267	computer games	computer games	100%
			Total 100%

Professional, Statutory and Regulatory Body (PSRB)
No PSRBs have been recorded

	The course involves a broad range of teaching techniques, including lectures, seminars, tutorials, computer laboratories, group exercises, case studies, oral presentations, practical work, industrial placement, and a substantial project in the Stage 4 (Honours Year). Lectures are supported by a range of equipment including whiteboards, video, and computer projection facilities where appropriate. Ample opportunity exists for questions, interaction, and discussion. The format is selected based on the suitability to the subject. It may consist of the staff supporting the student engage in problem-solving or may exist as a series of group exercises where each group is encouraged to allocate responsibilities, etc. Generally, teaching is used to discuss the underlying theories, support the peer-learning, clarify the material and experiment with the techniques and skills discussed. This may include the use of online platforms to bring interactivity and immediacy to the teaching and learning process, using platforms such as MentiMeter for quizzes, polls, word clouds, boards, and Padlet, or Office 365 tools for collaboratively creating boards or presentations. The ability to work as an
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What the Course Involves	effective member of a project team is an essential skill for anyone working in industry, and group working is a major feature of the course in Stage 3 (Degree Year). The nature of the Games and Computing subjects requires that students gain practical experience in the computer laboratory, and most modules involve lab work on a PC or workstation. The School has several laboratories housing an approximate total of 300+ PCs, equipped with the latest versions of open source and commercial software for the exclusive use of Computing students. The computing lab is the venue where students, supported by staff, practise, and enhance their skills in the use of industry standard development environments, and sophisticated software applications. In each semester, students are expected to attend three to four hours per week per 15 credit module in lectures/tutorials/laboratories, with slightly more time being allocated in the early stages of the course. Delivery follows the University Academic Calendar, with timetabled sessions during the teaching weeks. Students are expected to continue their learning through independent study activities outside timetabled sessions. Each semester includes a Connect and Reflect Week, which is not used to deliver new academic content, but to provide opportunities for students to consolidate learning, reflect on progress and development goals, and have a concentrated focus on learning and development priorities. This course offers the opportunity for students to tailor their experience through themed pathways by elective choice, introduced in the common foundation year at Stage 1, and expanded upon from Stages 2 to 4. During Stages 2 and 3, the student will study two core Computing modules, and four subject modules that are core to the course. The student will elect to study a subject core module from a different course within the School, to gain an appreciation for the wider context within which Computing operates and diversify their skill set. Uniquely, in Stage 3 (Degree Year), students will explore design principles and processes through engaging, fast-paced, interdisciplinary design projects, with further collaboration appearing earlier in the course to embed and support learning with a focus upon design, and later in the course through collaborative project supervision to support in diversifying skills via a Capstone Project. The Capstone Project is an extended piece of work, where students will be supported by Academics from multiple disciplines, to undertake the development of an artefact which will be reported on, relevant to their course. The student agrees a requirements specification for a task (which may be the development of game etc.) and then, produces a thesis which documents the research and development of an artefact relevant to the Games Design and Development course. The optional Short Industrial Placement or Year-long Industrial Placement are useful supplements to the other teaching and learning experiences. The interaction with the staff of the company, the required tasks and the general environment provide an experience difficult to achieve by other means. Students are supported on the selected placement by a Placement Supervisor, identified from staff within the School, and an Industrial Supervisor from the company and the School Placement Coordinator supports and monitors the assessment process. Students also have the option available (at the relevant interval within the programme) to participate in the Study Abroad scheme which allows studies to be engaged with at a suitable partner institution or work abroad within an appropriate industrial placement.
UCAS Code	Code GW12
Course Collaborations	
No collaborations listed	
	In today's gaming industry, maintaining and enhancing a competitive edge is a challenging endeavour. Success hinges on the ability to efficiently and securely access, process, and communicate information. The key to this success lies in the effective use of computing and creative design technologies. The gaming industry and beyond this, the wider computing and creative design landscape, is in constant need of skilled graduates capable of producing robust, user-friendly

Introduction to the Course	applications and developing cutting-edge computer software that seamlessly integrate within modern businesses, and solutions that integrate within specialist creative industries. Thinking creatively, developing critical design skills, reinforcing problem solving skills and exploring novel methods are just some of the features of skill toolsets our Games Design and Development graduates exhibit. Thriving in this dynamic and competitive field demands a distinctive blend of technical prowess and creative ingenuity. Our Games Design and Development course is meticulously structured and developed to equip students with a spectrum of specialist development skills and knowledge, highly sought-after in a rapidly evolving, fast-paced industry. This is a 4-year Honours Degree course, with the choice to embark upon an optional Year-long Industrial Placement or Study Abroad opportunity, which would extend the course to a 5-year programme. The course also offers a Short Industrial Placement option, for those wishing to explore shorter placement options during the Summer months. Such opportunities provide students with experience studying in another country or developing knowledge and skills in a real work environment. Students will acquire a comprehensive, hands-on understanding of the latest advancements in computing fundamentals within an applicable games context, and in turn, design, develop and deploy efficient and engaging game solutions. Students will cultivate both theoretical insights and practical expertise in applications across various platforms, spanning desktop and mobile platforms. This knowledge will be applied to tackle real-world commercial challenges, resulting in the creation of interactive games and applications tailored for a chosen platform with user interfaces optimized for the target audience. Additionally, students will be guided towards active engagement with ethical considerations throughout design processes. They will critically evaluate the potential impact of their games on players, society, and culture. Emphasis is placed on fostering responsible and inclusive design practices, ensuring that games are developed with sensitivity to diverse perspectives and without perpetuating harmful stereotypes. Through thoughtful discussions and assignments, students will explore the ethical dimensions of game content, mechanics, and monetisation models, ultimately aiming to create experiences that contribute positively to the gaming community and broader society.
	Overall Aim of the Course Graduates of the BSc (Hons) Games Design and Development course will have: • An understanding of the core concepts of the design and exploitation of computation and computer technology for Games Design and Development, including the importance of creative development strategies, design methodologies, marketing techniques, and innovative use of novel technologies. • The ability to apply disciplined critical design analysis and appraisal approaches to game solutions alongside the dissection of practical problems within computing. • The ability to strategically appraise appropriate target platforms, including an appreciation of the uncertainty, ambiguity, and trade-offs inherent in the deployment of code and associated media assets on these platforms, including cross-platform development approaches. • An understanding of relevant theory and analysis within computing in support of more expansive problem-solving methods specifically targeting game design solutions, including the ability to develop and use theoretical models to abstract and simplify complex processes. • The ability to recognise the legal, social, ethical, and professional issues involved in computing and creative subjects, and be guided by the adoption of appropriate industry standards and professional practices. The skills base of this course generally resides within the body of knowledge defined by the QAA Computing benchmark statement and the ACM Computer Science Curriculum . As well as the technical focus, the course contains content that prepares students in social, ethical, legal, and professional aspects of a cooperative human environment such as the workplace.

Educational Aims of the Course

This course offers the opportunity for students to tailor their experience through themed pathways by elective choice, introduced in a common first year (Stage 1), and expanded upon from Stages 2 to 4.

During Stages 2 and 3, the student will study two core computing modules, and four subject core modules (modules core to the Games Design and Development subject) throughout the year. The student will then have the chance to study a subject core module from a different course within the School of Computing, Engineering and Technology, to gain an appreciation for the wider context within which computing operates and diversify their skill set.

At Stage 4 (Honours), the student will study three subject core modules in the first Semester, and then select a subject core module from a different course as the fourth. In the second Semester, students will embark upon a final Capstone Project, culminating their skills in the final project.

The course aim is further expanded into Stage aims as follows.

Stage 1 (Certificate of Higher Education Computing)

- To embark upon a common first year, that will allow understanding of the breadth of computing as a multidisciplinary subject, with particular focus within the chosen course.
- To acquire knowledge and understanding of the fundamental technical concepts of programming and databases, which include the establishment of problem-solving skills within Computing, creative design fundamentals and software development practices using appropriate tools and approaches.
- To employ further skills, knowledge and understanding that promote continued personal and professional development through investigative activities that use a combination of written and computational work, both on an individual and group basis.

Stage 2 (Diploma of Higher Education Computing)

- To adapt skills and show knowledge and understanding within experimental game mechanics and contextual programming, with an appreciation for variable development pipelines.
- To undertake deeper consideration within the scope of game design and gameplay fundamentals and expand on critical appraisal skills.
- To practice knowledge and understanding with regards to the wider scope of Games Design and Development and importance of applied theory within gameplay and experiences.
- To distinguish and plan appropriate quality assurance mechanisms within a game context and, the wider field of computing.
- To appropriately adapt the application of technology to solve real-world problems given knowledge and understanding of the professional skills, and legal, ethical, and social issues involved.
- To express fundamental skills in prototyping, evidencing problem solving skills and implementation within a game engine environment.

Stage 3 (BSc Games Design and Development)

- To organise and formulate in an effective manner with colleagues and assume an active role in the planning and development of media artefacts, appropriate within a Games Design and Development context, both as an individual and as an effective member of a small project team.
- To assemble and make judgements on artefacts that manage information effectively using an appropriate range of media techniques, factoring in immersive technologies, 3D modelling and further innovative/emergent practices.
- To utilise a range of tools and techniques to develop systems with due regard to utility, accessibility, and usability.

	- To demonstrate an appreciation for the wider scope of games, including gamification concepts for non-game artefacts and practices involved surrounding such examples. Stage 4 (BSc (Hons) Games Design and Development) - To illustrate a wealth of practical research and development activities that support in the design and implementation of flexible creative computing artefacts, and within the context of games, consideration of users/players, tasks/requirements, AI for games, and the overall game experience. - To operate an appropriate range of tools and technologies, demonstrating an ability to plan, organise and implement a computing project with due regard to resource constraints, experience, human factors, client requirements and professional standards. - To develop effective, interactive artefacts that utilise and integrate a multitude of media-based technologies. - To illustrate in-depth knowledge and understanding of advanced and recent and emerging topics relevant to the game landscape. - To compose creative Games Design and Development solutions to complex problems through the synthesis of existing knowledge of technologies with state-of-the-art techniques identified from research of current literature across multiple academic subjects. - To critique different solution strategies that could be applied to a large-scale problem while being able to defend and justify the selected approach.
Awarding Institution	Robert Gordon University
Course Duration	Blended Learning Full-Time - The course will normally be delivered and assessed over the following duration(s). Please note the period from enrolment to final assessment will depend on the point of intake and scheduled University breaks. Blended Learning Full-Time: 4 to 5 years, including an optional Year-Long Industrial Placement in-between Stages 2 to 3, or Stages 3 to 4. Advanced Entry: The course provides opportunities for articulation into Stage 2 (with a relevant HNC or equivalent) and Stage 3 (with a relevant HND or equivalent). Advanced entry into Stage 2 results in a course that is 3 to 4 years, including an optional Year-Long Industrial Placement in-between Stages 2 to 3, or Stages 3 to 4. Advanced entry into Stage 3 results in a course that is 2 to 3 years, including an optional Year-Long Industrial Placement in-between Stages 3 to 4. Advanced entry into Stage 3 will be subject to conditional/fixed elective modules or similar depending on subject area and this will form part of the course induction process. Exit awards are attainable at the end of each Stage.
	The BSc (Hons) Games Design and Development course includes subject material that explores an exciting blend of computing, games design and applicable development practices that will enable students to develop sophisticated applications and artefacts using technical skills and creative fundamentals.

Distinctive Features of the Course	The course is taught by a team of Full-Time and Part-Time staff, including several highly distinguished active researchers in the fields of computing, graphics, animation, immersive technologies, games and gamification. Several of the teaching staff have extensive experience of applying media technology in commercial and educational applications in addition and in tackling industrial challenges. The course enables the students to benefit from their experience and involvement in the leading of innovative development and application of game design fundamentals and developmental technologies to real problems applicable to the games domain. The optional Year-long Industrial Placement between Stages 2 and 3, or Stages 3 and 4 of study, provide students with the opportunity to gain the invaluable experience of working in a real business environment with professionals, dealing with real users, clients, and their problems. For students wishing to take on a shorter placement, there is the option to take a Short Industrial Placement during the Summer months in-between Semesters, again between Stages 2 and 3, or Stages 3 and 4 of study as indicated in the Course Structure. The course also provides the students the option to Study Abroad between Stages 2 and 3, or Stages 3 and 4 of study. The School of Computing, Engineering and Technology partners with several European institutions and has recently participated in exchanges with Belgium, Finland, Norway, and Sweden. Stage 1 of all Undergraduate courses within the School run with a common first year as the core focus on the course subject, allowing students to understand computing as a multidisciplinary subject and build their community of practice across the School. Expanding on this, the course includes the opportunity to further diversify on skill set through elective choices in each Semester (except Semester 2 in Stage 4, where the Capstone Project resides) through each Stage of the course (Stages 2 to 4) which in turn, provides students with core and subject core modules, and the opportunity to select a subject core module per Semester from a different course within the School. This allows students on the course to expand their skill set and pursue various themes of interest that compliment Games Design and Development. During Stage 3 (Degree Year), students develop their project skills and interdisciplinary group-based learning further alongside understanding about the concepts and principles of experiential development within games and through a broader lens. In Stage 4 (Honours) , students will focus on subject core, cutting-edge topics, and with the year culminating with the final Capstone Project, where students will be supported by Academics when looking to undertake a substantial and impactful project, and that harnesses the various competencies explored throughout their degree to further reinforces desirable skills applicable to the chosen sector.
	Students may elect to undertake a short summer placement in a summer break and/or a year-long placement between stages. The placement provides an opportunity to sample working life, be it in commerce, industry, or the public sector, and apply knowledge and skills learned in the classroom to the challenges of the workplace. Students who have completed the placement as part of their course have a definite advantage when it comes to applying for graduate positions. With the combination of training, experience, and a successful degree, students are often offered subsequent employment by the placement provider with which they undertook their placement. While the student is on placement, they will be responsible to a designated member of the placement provider (the Industrial Supervisor) for the discharge of their day-to-day duties. Each student on placement is also assigned an Academic Tutor from the staff of the School, who will act as the student's personal tutor during the placement. The Academic Tutor will hold a small number of scheduled video calls with the student and their Industrial Supervisor during the placement. Students may choose to undertake a short summer placement between Stages 1 and 2, or Stages 2 and 3, or Stages 3 and 4. This normally comprises at least 4 weeks of continuous,

Placement/Professional Experience	full-time supervised work experience. The assessment for a short summer placement experience is based on a review of the student's work-related skills, conducted by the Industrial Supervisor. Students on a short summer placement will normally receive at least 1 video call with their Academic Tutor. In addition, or as an alternative to a short summer placement, students may undertake a year-long placement either between Stage 2 and 3, or between Stage 3 and 4. The year-long industrial placement normally comprises of 12 months of continuous, full-time supervised work experience. The assessment for a year-long placement is based on a review of the student's work-related skills, conducted by both the student and their Industrial Supervisor. Students on a year-long placement will normally receive at least 2 video calls with their Academic Tutor. The University provides support to students interested in undertaking a placement through the Placement Office and Centre for Employability and Community Engagement. The Placement Office maintains a pool of placement opportunities and establishes contact with new placement providers as necessary. It will normally assist students by informing them of suitable placement opportunities. Students are also given help and guidance by the Placement Office in obtaining their own placement from those advertised directly by companies or through their own connections. The Centre for Employability and Community Engagement offers further support for students, including with preparing their CV and developing interview skills. All placements require approval from the School prior to commencement, to ensure that they are suitable for the student and to confirm that the placement provider has put adequate supervisory and health-and-safety arrangements in place. Placements cannot be guaranteed and students wishing to undertake a placement are expected to reasonably engage with the process, for example by providing a copy of their CV to the Placement Office and by attending any interviews with potential placement providers when required. The credits awarded for the placement module are in addition to the 120 credits achievable from taught modules at each stage. Successful completion is not required for progression to the following stage. The credits achieved for placement may not be used to compensate for any credits from the taught modules.
Opportunities for Further Study	An Honours Degree in Games Design and Development attained with at least an Upper Second (2.1) Classification will normally provide access to Postgraduate and Doctoral programmes in Higher Education Institutions.
External and Internal Indicators of Quality and Standards	QAA (Quality Assessment Agency for Higher Education) published Computing Benchmarks (2000, 2007 and 2015) outlining the standards expected for the award of Honours Degrees in Computing. QAA Benchmark Statements are used during the design of modules, courses, and the Undergraduate Computing Programme. The structure and content of the course has been constructed following the QAA's Computing benchmarks and the QAA's principles of course design (both the current benchmarks and those described in the 2015 Benchmark Statements were consulted). The course design has been strongly influenced by three key computing concepts: computational thinking, the process of analysing and developing computer systems, and the balance between theory and practice. The content of the course has been selected to provide a balanced coverage of the three broad categories of abilities and skills identified in the QAA Computing Benchmarks: computing-related cognitive abilities, computing-related practical abilities, and additional transferable skills. These have been formulated as learning outcomes within the current Scottish Credit and Qualifications Framework (SCQF) . SCQF credit values have been used as a guide during the design of modules while the structure and content of the course has also been designed with the British Computer Society's (BCS) requirements for Practical/Engineering Applications in mind and embedded in the core structure are modules and topics which provide students with computing skills and knowledge in Underlying Theory, Design and Methodology, Tools and Techniques and Project Management and Group work. Standards are primarily established through QAA (Quality Assurance Agency) Benchmarks and the requirements imposed by the British Computer Society to gain professional course recognition. The achievement of appropriate standards is demonstrated through the University's course validation procedures and professional institution accreditation visits. The QAA qualification descriptors and Subject Benchmark Statements

	provide a reference framework that has guided the evolution of the School's programme, and the extent of the subject discipline, as defined by QAA, is broadly matched by the provision that now exists within the School's undergraduate programme. Several internal processes and procedures are used by the School to ensure that standards are maintained and enhanced. Stakeholders' views on standards are routinely obtained from the Staff/Student Liaison Committee, Undergraduate Programme Management Team, Student Experience questionnaires, Industrial Board, industrial visits to placement students, staff involvement in professional body activities and staff participation in accreditation events outside the School and at other HE institutions. In addition to this, best practice in teaching and learning within the University is monitored through the University quality enhancement and quality assurance processes. These include the regular five-yearly cycle of Institution-Led Subject Reviews, as well as participation in the University quality assurance committee structures, such as the University's Student Experience, and Academic Regulations Sub-Committees. Finally, it should be mentioned that the lecturers within the School have an ongoing interest in pedagogical research and a commitment to expanding the scholarship of computing science education. Academic staff help to organise, and participate in, national and international conferences and have been awarded funding from the Higher Education Academy, the Scottish Informatics and Computer Science Alliance, and the Scottish Enhancement Themes to pursue work in computer science education, as well as more general work on student transitions. The School also participates in university conferences on teaching and learning.
University Academic Regulations	This course is governed by the provisions of the University's Academic Regulations, which are available at www.rgu.ac.uk/academicregulations . In particular: Regulation A1: Courses Regulation A2: Admission and Enrolment Regulation A3: Section 1: Student Appeals (Awards and Progression) Procedure Regulation A3: Section 2: Student Conduct Procedure Regulation A3: Section 3: Student Fitness to Practise Procedure Regulation A3: Section 4: Student Fitness to Study Procedure Regulation A4: Assessment and Recommendations of Assessment Boards
University Entry Requirements	Applicants must satisfy the University's general admission requirements to courses as contained in Academic Regulation A2: Admission and Enrolment, including proficiency to a minimum standard in the English language. Specific entry requirements for this course are detailed below.
Course-Specific Entry Requirements	Minimum Qualification Requirements Scottish Higher (BCC - CCCC), to include English and Maths at National 5 grade C or above if not held at Higher. Standard Qualification Requirements Scottish Higher (BBCC), to include English and Maths at National 5 grade C or above if not held at Higher. A Level (CCC), to include English and Maths at GCSE grade 5 if not held at A Level. International Baccalaureate Diploma Programme (26 points), to include a pass in Standard Level English and Maths at grade 4 or above if not held at Higher Level. Leaving Certificate - Higher Level (Ireland) (first awarded in 2017 - H3, H3, H3, H3), to include English and Maths at grade O3 or above if not held at Higher.

	Applicants from partner Colleges for entry to stage three An appropriate SQA HND qualification in a relevant subject, for example; HND Interactive Multimedia Creation, HND in Games Development or an equivalent two-year Higher Diploma. All other applicants who are interested in applying for advanced entry will be considered on an individual basis. Applications from students with non-standard qualifications or work experience will be considered on an individual basis. For general information on entry requirements, consult the RGU website - likewise for specific details on qualification outcomes: https://www.rgu.ac.uk/study/apply/entry-requirements-explained .
Alternative Qualifications for Entry	Equivalent qualifications, e.g SQA, BTEC, S/NVQs, overseas qualifications and Access Programmes will be considered on an individual basis. Applications from mature students with non-standard qualifications or work experience are welcomed.
Interview Procedures	There is normally no requirement for prospective applicants to be interviewed.
Course-Specific Regulations	Honours Classification is based solely upon the student's performance in Stage 4 of the course. Credits achieved for the short term Summer Placement and/or the Industrial Placement module cannot be used to compensate for credits not achieved in other modules of the course. The Capstone Project is deemed to be central to the aims and objectives of the course and therefore, cannot be condoned. Electives will run at the discretion of the School. Students cannot choose an elective module that they have already taken on the course, for example, some electives are aimed at particular cohorts such as articulating students. Some electives will not be available until the module is first delivered by the School, therefore, please consult the Course Leader for further information on elective availability
Equality, Diversity and Inclusion	The University has a strong commitment to the active promotion of equality across its functions, including its course provision. To this end the University has developed an inclusivity statement which articulates the University's desire to strive towards creating a working, learning and social atmosphere which is inclusive, harmonious and respectful of diversity. The University has also developed an Equality and Diversity Policy and every two years publishes a mainstreaming report, including equality outcomes, which explains how the university is developing and implementing its inclusivity statement. At a course level, Course Leaders monitor and comment on equalities data through the Annual Appraisal process. Through this process any significant trends in equalities data are highlighted. Further information can be found at: Equality & Diversity RGU.

	All Schools have a nominated School Contact who can advise on the requirements of the course and the range of course-specific reasonable adjustments that may be made. The University's Inclusion Centre provides support for students, details of which may be found at: The Inclusion Centre: Disability & Dyslexia Support & Advice Services RGU.
University Strategy	The Purpose of the University is: - To be an innovative, inclusive and professionally focused University that has a positive impact on those we serve. The Mission of the University is: - To transform people and communities by providing excellent teaching, research and enterprise opportunities to contribute to economic, social and cultural and environmental sustainability. The Culture of the University is: - To recognise and celebrate the values of curiosity, authenticity, approachability, collaboration, enterprise, ambition, respectfulness, and equality. Driven by the spirit of curiosity to continually improve, our values and principles will guide us. Our partnerships with other organisations and the communities we serve are also a mechanism to develop and grow our diversity. We have a strong commitment to widening and extending access to our courses, regardless of an individual's background. At the core of the University's significant achievements is the commitment and dedication of our people. The sense of partnership between our staff and students is strong and supporting our students to succeed in their studies and beyond is deeply ingrained in our culture. These features are reflected in the University's student satisfaction rates, which in recent years have consistently placed RGU in the top 10 of universities in the UK. RGU has a long history of inspiring highly employable graduates. This is achieved through close, collaborative contact with employers in the design and delivery of our curriculum; the range and breadth of accreditations with professional and statutory bodies; the significant role work placements and work-based learning play within our courses; and the embedding of enterprise across all our activity. In a changing society, we will ensure the continued evolution of our curriculum so that our graduates can continue to succeed in their ambitions and make a positive impact within society well into the future. To achieve our ambitions, our curriculum must be global in outlook, and we must maximise the international reach of our course portfolio. The University's research activities must have international impact and our enterprise activities must reflect the global reach of the economic sectors we work with and the ambitions of the region. RGU recognises that the creation of a sustainable future for all is a priority that unites us globally, and which demands expertise, creativity, and innovation from all subject areas and professional disciplines. Consequently, sustainability is integrated into all courses, equipping students with skills and competencies that are essential for their future careers and lives.
	Philosophy Of Course Approach To Learning The BSc (Hons) Games Design and Development course is an exciting programme of study that explores fundamental design techniques, game theory, concepts of gamification, prototyping, and the wider areas of interactive development. As standard within School's courses, the Games Design and Development course is underpinned with a set of core modules such as programming and professional skills, to create

Course-Specific Learning Experience

diverse graduates with flexible, transferable skills. Students opting to study on the Games Design and Development course will build upon any (where applicable) existing skill set, aiming to further develop knowledge and skills in subject areas associated with concepts of critical design. Students will explore and discover relevant emergent technologies to effectively develop systems/artefacts that explore rapid prototyping and responsive design, information visualisation and accessibility, games and gamification, and immersive technologies, to name but a few.

This in turn will provide relevant professional industries with diverse, skilled practitioners in various fields of Games Design and Development, such as creative computing, user centred design, 3D modelling and visualisation, and of course, games and gamification. Using project based learning, the course will also provide opportunities for students to select and study specialist topics, fitting within the context of Games Design and Development.

It is expected that students engage with, and actively participate in the learning process throughout the course, developing their skills and knowledge to a high level. Regular attendance and engagement with all course components are a mandatory requirement to achieve a satisfactory level of performance and, to ensure full advantage of the opportunities available are taken.

There is a strong effort made with the integration of appropriate topics within computing, delivered by staff with a background in applied research, industrial experience, and innovation in the application of learning technology to novel teaching methods.

Innovation Through Investigation and Understanding of Computing Technologies

The ubiquitous and all-pervasive nature of information technology has had a profound effect on modern life. Computers and media-based systems are becoming key components of an ever-increasing range of products and services that are both an integral part of everyday life and key components of business, industry, private and public life. The scale of computing extends from the smallest microprocessors embedded in smart cards and mobile phones to global networks that can harness grids of PCs to solve problems that demand super-computer performance. Computing and Games Design and Development professionals play a crucial role in developing this technology and applying it in innovative ways to almost all areas of business and public sector organisations. Increasingly, the information handled and delivered by computers is represented by a variety of interactive media. The challenge is to develop the right product, using the right combination of media type, for the right people. Innovation alone is not enough; the successful use of Games Design and Development techniques rests on an ability to research and investigate the real requirements of the organisations and users who stand to benefit from innovation.

Professional Skills

Professional skills, such as project management, communication, presentation, and writing skills are extremely desirable in the creative and wider industries. Several modules over the course provide students with skills in these areas and opportunities to demonstrate them, so they are fully prepared to enter the workforce on graduation. The course culminates with the Capstone Project, where each student must deliver a written thesis and present a live demonstration of their work. This allows the students to show a full complex piece of work they have designed, documented, and implemented and that they can be proud of producing.

External Competitions and Challenges

As part of the course, the students are encouraged to take part in various competitions and challenges throughout the year.

Such challenges include:

- **Hackathon** (RGUHack), where students work in teams for approx. 24 hours on challenges set by industry partners.
- **Game Jam** (RGU Game Jam), where students can work individually or in teams for approx. 48 hours on various digital and tabletop game challenges whilst being mentored by industry partners and Academic staff.
- **The Google Hash Code challenge** , where the students work on large optimisation challenges set by Google over a period of four hours.
- The School has also sent teams of computing students to compete in the **UK Cyber Security Challenges** .

The Course Team strongly believe, and student feedback has shown, that competing in these events raises students' confidence in their own ability and allows them to use the knowledge and skills they have learned in a pressure, but consequence-free, environment.

Industry Involvement

The course prides itself on its close links to industry, as part of this, some project modules feature representatives from industry partners acting as problem holders that the student will aim to design, develop, and test solutions for. There are also several opportunities for industry partners to visit the School and host visits on their premises for students, and in addition, offer said partners to become involved in the **School of Computing, Engineering and Technology's Industry Board** .

Technical Facilities

Many of the practical aspects of the course take place in the computer laboratory. The School has several laboratories housing an approximate total of 300+ PCs, equipped with the latest versions of open source and commercial software for the exclusive use of computing students. The School also has a range of additional hardware for use in teaching, including iPads, Raspberry Pi devices, photography equipment, and virtual reality headsets. Throughout the School, there wi-fi coverage that allows access to the University network. The School's labs are regularly updated on a 3-year replacement cycle to ensure that the School's facilities can deliver high performance and can run the latest software. Access to the labs is by electronic key fob at any time during University opening hours. Software provision is comprehensive, including compilers for most modern programming languages, database systems including Oracle, SQL Server, and MS Office 365 packages. On the School's PCs, standard installed software includes (amongst others) Microsoft Office 365, Adobe Creative Cloud and Production suites, Java, VS Code, Git, NodeJS, and Android development tools, alongside Unity Game Engine and specialist animation/3D software such as Toon Boom Harmony and Blender. Much of the software used across the course is flexible and open source, or can typically be acquired at low cost for students via educational licensing options. This allows students to continue their work at home using a similar set up, available in the labs and also - transfer their knowledge across several solutions where appropriate.

The Learning Process: Teaching Strategy and Methods

The course curriculum is designed with an emphasis on the integration of theory, practical skills, and application, with an emphasis on a truly interdisciplinary experience and diverse skill building. It enables the student to acquire and develop the subject knowledge and understanding necessary for successful professional employment and supplies methodologies for the adoption of a disciplined approach to games design development tasks such as problem analysis, specification, critical design, implementation, and testing.

Such knowledge includes an understanding of the range of possible options for these tasks, suitable technologies and media types, the related practical and transferable skills including relevant approaches to collaborative work, access to the appropriate software tools and resources. It also provides the necessary underpinning to guide practice, to ensure the sustainability of their knowledge and to provide an appropriate framework that will accommodate rapid technological change and advancement.

These skills are developed more intensively at the Stage 3 (Degree Year) and Stage 4 (Honours Year) levels corresponding to the Scottish Credit and Qualifications Framework (levels 7 to 10). As the student progresses through the course they are encouraged to expand their understanding and critical appreciation of Game Design and Development, and to extend skills in problem analysis, specification, critical design, implementation, and testing alongside maintenance into a multitude of artefacts.

A general feature of the course is that it stresses a unified, integrated view of the subject area. This is reflected in the fact that it is designed to:

- Present a coherent underpinning of the theory appropriate to the aims of the course which is further developed and used throughout the period of study. This enables graduating students to adapt to future developments in the field.
- Reflect the rapid rate of change in the field and ensure that coverage is given to a selection of emerging topics so that students are aware of likely future developments in the subject, together with their potential impact.
- Integrate the delivery of teaching across topics to highlight fundamental aspects of transferable knowledge, interconnectivity and general overlap within different areas of the syllabus.
- Integrate modes of assessment to attempt to balance the types (individual and group) and purposes (e.g. formative and summative) of such activity.

- Facilitate students to exhibit competence and achievement across several different topics (integrated assessment).
- Encourage students to combine elements of their learning from different parts of the programme and to show their accumulated and integrated knowledge and understanding of a topic or subject area (synoptic assessment).
- Obtain general and specific skills from the professionally orientated nature of the course, being addressed by a combination of teaching and learning methods, including but not limited to; lectures, tutorials, computing laboratories, workshops, seminars, practical work, industrial placement (where appropriate), integrative project work, student-centred learning, and private study.

Teaching and Learning Approaches

Lecture

This is usually a formal discourse for the purposes of dissemination of information, the demonstration of techniques where appropriate and the discussion of supporting ideas and consequences. The lecture is supported by a network of technology assisted dissemination equipment including computer projection facilities, internet resources, media resources and whiteboards/presentation tools where appropriate. Although this type of presentation is suitable for a one-sided discourse, ample opportunity exists for questions, interaction, and discussion - and this is facilitated by technological means, e.g. using personal response systems. In many of the modules taught on this course, lecturing occurs in a more interactive format, during laboratory sessions for example, using a studio lab or language lab approach where 'bite-size' lectures with theoretical content are interspersed with practical student-centred skill-based activities to embed theory in practice, encouraging deep learning.

Teaching and Learning Strategy

Seminar and Tutorials

In the School, tutorials are used for a wide range of activities, each suited to the subject. They may consist of the staff supporting students engaged in problem solving or may exist as a series of group exercises where each group is encouraged to distribute responsibilities allocate people to tasks and manage workload to complete some team-based activity. Generally this type of teaching is used to support the lecture, clarify the material and experiment with the techniques and skills required.

Laboratory

The nature of Games Design and Development, and the principles, theory and practice which make up the subject, requires students to gain practical skills in the use of computer technology. Laboratory activity takes place in one of the School's computer laboratories and consists of the student, supported by staff members - practising those skills necessary in the development and deployment of game-based artefacts such as prototypes, and including quantifiable analysis techniques for testing activity such as playtesting. There has been increasing emphasis on using a 'language - lab' approach to subjects in which a series of short mini-lectures are delivered in labs to allow students to gain theoretical knowledge of a topic. These are interspersed with selected practical exercises, which allow this knowledge to be immediately assimilated and so reinforce the learning process.

Industrial Experience

The course includes optional placements both on a short or year-long term, which act as a useful supplement to the other teaching and learning activities of the course. Students who wish to participate in an available, optional placement are required to apply competitively by submitting an appropriate cover letter and CV, and by attending interviews. Students on placement perform in a variety of roles and undertake a range of tasks in the companies for which they are working. They are supported by a Placement Supervisor, drawn from the Academic Staff within the School, and an Industrial Supervisor from the company in question. The interaction with staff in the company, the required tasks and the general working environment provides an experience of a suitable profession, appropriate to the Game Design and Development course, which is difficult to achieve by other means. The **School's Placement Coordinator** supports and monitors this process.

Group Projects

Group projects are extended pieces of work which are usually completed under supervision of a staff member. Students are assigned to small project teams and given a requirements specification for a task which may comprise elements of analysis, design and the implementation of an interactive system, development of a piece of software, or the exploration and

practical investigation into other game related activities. The students manage the project processes themselves and work as an organised team to produce a set of deliverables which usually includes a design and development component incorporating an interactive media-based artefact. In addition, they would be expected to provide accompanying documentation to a professional standard. This activity approximates closely with the development of a project undertaken by a project team within industry.

Individual Project

This is an extended piece of work, completed under supervision of a staff member. The student is expected to agree a requirements specification for a task, which may be the analysis and design of a system, the development of software or construction of a digital media artefact, and the production of a written project report. This activity approximates closely the development of an extended individual development task within industry.

Student's Role in Learning

Students are expected to be active participants in the learning process. This means that there is a requirement for students to extend their learning behaviour from reliance on formally delivered materials to wider reading, research, and self-directed enquiry. As students develop themselves over the duration of the course, there is an expectation that they become increasingly self-directed and self-motivated, taking greater control of the study process, and defining their personal objectives for learning. Students are encouraged to regularly reflect upon their learning and to establish a view of their own progress and development, which may form the basis of discussion with staff who provide guidance during tutorials. Students are expected to utilise the full range of facilities and resources available to them, for example the services and resources provided by the library.

The Assessment of Student Work

Assessment in learning is a key part of the pedagogical approach used within the School. Transparency of details and clarity is taken very seriously, and the School Office underpins this by preparing a schedule for both staff and students, outlining an assessment timetable to ensure workload and deadline tensions are suitably managed.

Assessments have been developed so that they may ensure module and course-wide learning outcomes have been met following the SCQF Level 7-10 characteristics of knowledge and understanding; practical skills; generic cognitive skills; communication, numeracy and IT skills, autonomy, accountability and working with others.

Communication is provided upon the finalisation of this to ensure students understand what assessment requirements they should be considering, and the criteria in the context of coursework and grading structures. Assessment is carried out in accordance with University Regulations and makes use of the University Grading Scheme, details of which can be found in the relevant sections of the University Academic Regulation [A4: Assessment and Assessment Boards](#).

It is important to note that all grades distributed to students during the session are provisional until the deliberations of the Assessment Board for the course. Provisional grades are presented at the Board, which consists of academic staff involved in course delivery as well as External Examiners. A final decision is confirmed on the performance of every student for each module undertaken and this decision is subsequently relayed to each student in an appropriate form. It may be accessed electronically on the Student Portal (My Gateway).

Any potential barriers to students with disabilities or specific needs are assessed by the **School's Support Team** who then liaise with appropriate teaching staff to ensure that such students are not disadvantaged during an assessment. In the case of examinations, the University has specific policies to which the School adheres.

Assessment Types

Many modules use a combination of assessment types that integrate modes of assessment to balance the types of activity (e.g. individual or group activities) and purposes of assessment (e.g. formative or summative). Where several modules cover allied subject material (e.g. modules in technical areas which are studied in conjunction with project management topics) various forms of integrated assessment may be used to promote the demonstration of proficiency across several different themes and transferable skills.

Assessment

In all years, synoptic assessments embedded into projects and other practical assignments are used to encourage students to combine elements of their learning from different parts of the programme and to show their accumulated knowledge and understanding of a set of topics or an extended subject area. A variety of assessment methods are used in this course, as deemed appropriate to the subject material and topic being assessed, and the context in which the material is delivered.

A student's Honours Classification (First Class Honours, Upper Second Class Honours etc.) is based solely on performance in Stage 4 of the course and ultimately, the **capstone project**.

Work Based Learning

Optional placements are assessed by a combination of Tutor and Supervisor input based on a form of Learning Contract, supported by a Learning Log.

Computer Aided Assessment

Computer-assisted assessment packages, such as the **quiz facility** on Moodle, are used in many modules, both for formative and summative assessment, to enhance student engagement and increase the speed of feedback.

Peer Assessment

Peer Assessment is utilised in many modules throughout the course where group work is undertaken. This serves to develop team working and other interpersonal skills, as well as enhancing appraisal and evaluation skills. Peer assessment may be used in both summative and formative contexts.

Practice Exams

Practice exams can take on the form of both practical (applied) and non-practical (documented) approach. This is typically a formative opportunity to prepare for the summative assessment under simulated exam conditions.

Exams

In general, exams - referred to as **unseen exams** , are used to assess knowledge and understanding. These are summative assessment points and are held under real exam conditions.

Assessed Laboratories

Assessed Laboratories are used to assess knowledge, understanding and, practical skills. They generally involve the completion by students of an unseen practical task such as writing software or developing a part of a system. They are usually held under supervised conditions with a fixed time constraint, but students may have access to supplementary material to help them complete the task.

Coursework

For most modules within Game Design and Development, summative assessment surrounding how the student has achieved the module's **Learning Outcomes** , is assessed by a combination of exam and coursework, with the latter being more of a focus. Coursework may take many forms, e.g. reports, essays, production of software, design documentation and modelling activities, game prototyping and testing etc.

Projects/Thesis/Dissertations

Extended projects are used in later Stages as a tool to allow students to tackle more complex problems that often require the integration of knowledge, understanding and intellectual skills from a variety of subject areas. These projects often require the practical application of several different technologies and provide students with the opportunity to learn new investigative and evaluative skills. Projects (particularly group projects) are also often used as the vehicle to develop valuable employability skills. Project work is usually assessed by an appraisal of the completed project deliverable, together with the submission of an accompanying report or dissertation which documents the work undertaken by the student.

University Support for Teaching, Learning and Assessment	The University provides a number of means of supporting teaching and learning: • student induction organised on a course or School basis; • an ongoing scheme of personal/pastoral support for students, including online support, provided by the <i>Inclusion Centre</i>; • an extensive programme of student study skills delivered through <i>Student Life</i> and the <i>Library</i>; • an extensive library of learning resources; • close collaboration with industry and professional, statutory and regulatory bodies; • the University's active participation in the Quality Assurance Agency's Quality Enhancement Themes, enhancementthemes.ac.uk ; • support for staff from the <i>Department of Academic Quality and Learning Enhancement (AQLE)</i>, including credit rated teaching, learning and assessment provision; • extensive opportunities for student placements with companies or organisations, academic overseas exchange programmes, and support for entrepreneurial activity and community engagement opportunities; • access to a range of professionally qualified, quality assured careers professionals to support individual students and graduates career planning and opportunity search, supported by a range of bespoke and contemporary online advice and information including the extensive RGU eHub; • Moodle, http://campusmoodle.rgu.ac.uk/ , the University's dedicated virtual learning environment; • a commitment to knowledge exchange and technology transfer through focused research activity, which contributes to the critical underpinning for all taught courses; • the expanding provision of state-of-the-art, purpose-built facilities and buildings.
Course-Specific Support for Teaching, Learning and Assessment	In addition to the University-wide support for Teaching and Learning, the School provides additional support for students enrolled on any of programmes taught. Members of staff publish course-based teaching and learning materials on the University's Virtual Learning Environment, Moodle . The range of published materials spans from supplementary teaching materials such as additional lecture notes and tutorial guides to software development to general guidance for students, such as codes of conduct in laboratories. Guidelines for course elements such as the capstone project module can also be found here. In addition to this, students can access study support material from the Department for the Enhancement of Learning, Teaching and Assessment (DELTA) . This is the central department that supports the School in implementing the university's strategic goals in teaching and learning, extending access and the student experience accessed here . Feedback is provided continuously to students through informal contact with Module Coordinators , Demonstrators and Lab Assistants in a classroom setting and feedback is provided on assessed practical work normally within 20 working days of the submission of work. Transcripts are made available electronically via the Student Portal following ratification by the Assessment Board. During Semester 1 and 2, there is the Connect and Reflect Week , which pauses teaching activity a few weeks into a Semester to allow students a chance to review and catch up on topics, advance their knowledge and also get involved in activities taking place during this week to reinforce their learning and social activity through extra-curricular events. The Student Handbook provides detail on the School, course, and working environment. It contains information regarding teaching and assessment, IT support, student support services, learning resources including the library and health and safety. It details key contact names and email addresses. The handbook is typically located on Moodle and will be organised into sections so that information is readily accessible.

University Monitoring of Quality and Standards	The University employs several mechanisms for evaluating and improving the quality and standards of teaching, learning and assessment, including: • <i>Annual Course Appraisal Reports</i> are prepared for each course and reviewed and approved by Course/Programme Management Teams and School Academic Boards which consider, amongst other things, feedback generated from student questionnaires; • <i>Institution-Led Subject Review</i> , involving external panel members, on a six-yearly basis to formally review its major subject provision, followed by a three year interim review to monitor progress against actions/issues raised through the review process; • <i>External Examiner Annual Reports</i> ; • Ongoing liaison with industrial/professional liaison groups. Formal Committees with responsibility for monitoring and evaluating quality and standards: • Staff/Student Liaison Committees, or equivalent; • Course/Programme Management Teams; • School Academic Boards; • Assessment Boards; • Quality Assurance and Enhancement Committee and associated Sub-Committees.
Course-Specific Monitoring of Quality and Standards	Annual Course Appraisals are reviewed and approved by the Joint Programme Management Team and School Academic Board, which meet at least once per semester. The Joint Programme Management Team considers all aspects of curriculum development as well as Teaching and Learning enhancements arising from feedback generated via External Examiner Annual Reports, Student Staff Liaison Committees (SSLCs), National Student Survey (NSS) and Student Voice Surveys. These enhancements are also discussed at length in regular meetings between the Dean of School and Course Leaders, as well as Whole School meetings. The Programme Management Team and School Academic Board also review all proposed changes to the structure and content of the Undergraduate and Postgraduate Programme, arising from quality assurance and enhancement initiatives at the university level. The Undergraduate and Postgraduate Programme is also informed by feedback from the School's Industry Board.
	The University and RGU:Union work in partnership to create an environment which stimulates: • the participation of students, by empowering them to proactively provide views and opinions; • a supportive learning community where students and staff engage in meaningful dialogue; and • the engagement of students in the design of solutions and enhancements. This approach is underpinned by the <i>Student Partnership Agreement</i> which promotes the ethos of partnership at all levels of the University. Further information can be found at the Student Representation and Partnership Moodle Area: Student Representation & Partnership RGU .

University Student Feedback Mechanisms	There are a variety of opportunities for students to provide feedback to the University, and to become actively involved in shaping their learning experience. These opportunities are integral to the University's approach to the quality assurance and enhancement of teaching and learning, and the holistic student experience. Mechanisms through which students are engaged and supported include: • participation in student evaluation questionnaires and where relevant the <i>National Student Survey (NSS)</i> ; • staff/student liaison arrangements at course/programme level; • support for Student Representatives; • student representation on Institution-Led Subject Review Panels and Validations; • student representation on key University committees.
Course-Specific Student Feedback Mechanisms	There is a Student Staff Liaison Committee (SSLC) for all Undergraduate Degree courses, alongside an additional committee for Postgraduate Degree courses. It normally consists of the Dean of School, Associate Dean for Academic Development and Student Experience and Course Leaders, with student representatives from each stage of each course. The Committee aims to be held twice a semester, feeding information on to appropriate Course Leaders and Module Coordinators, then onto School Committees as required. Student representatives also take part in the School Academic Board, where they can contribute to discussions about School policy. In addition, there are regular contact points with Course Leaders and student representatives to discuss issues arising within specific classes and on individual modules, in addition to negotiating any challenges and enhancements. Student Voice Surveys are made available to students on all modules each semester, which allows for the collection of feedback on student experience including enhancements and good practice so actions can be put in place where required, or more of what student's enjoy embedded into the teaching.

Awards

Stage 1

Permissible Awards	Title	Location	Target Award	PSRB	Award Credits
Certificate of Higher Education - CertHE	Computing	Aberdeen, United Kingdom	Exit Award		120

Stage 2

Permissible Awards	Title	Location	Target Award	PSRB	Award Credits
Diploma of Higher Education - DipHE	Computing	Aberdeen, United Kingdom	Exit Award		240

Stage 3

Permissible Awards	Title	Location	Target Award	PSRB	Award Credits
Bachelor of Science - BSc	Games Design and Development	Aberdeen, United Kingdom	Exit Award		360

Stage 4

Permissible Awards	Title	Location	Target Award	PSRB	Award Credits
Bachelor of Science with Honours - BSc (Hons)	Games Design and Development	Aberdeen, United Kingdom	Target Award		480

Level Learning Outcomes

On successful completion of this Course the learner will be able to :

#	Description	SCQF Characteristic	SCQF Level
LLO1	Demonstrate the principles and application of the concepts involved in the requirements specification, design, implementation, and testing of software systems that utilise media, interactive or otherwise.	SC1	
LLO2	Demonstrate the principles and application of the concepts involved in the use of media in applications, Human Computer Interaction and the perception and visualisation of information.	SC1	
LLO3	Demonstrate the role of computing and information systems within an organisation, the promotion of judgement and decision-making capabilities in selecting appropriate computer solutions to problems in organisations and individual users, and their specification for implementation where necessary.	SC1	
LLO4	Consider the issues involved in the management of information and technology in an organisation, and the development of industrial information systems that utilise digital technologies.	SC1	
LLO5	Analyse the requirements of users and organisations to design, implement, evaluate, and manage computing and media-based applications within the game domain.	SC2	
LLO6	Use appropriate tools to evaluate computing, games, and media-based applications in terms of general quality attributes and possible trade-offs presented within a given problem and context.	SC2	
LLO7	Deploy effectively the tools used for the construction and documentation of computer applications involving games design and development, with particular emphasis on understanding the pipeline and process involved in the effective deployment of interactive artefacts to solve practical problems.	SC2	
LLO8	Work as an effective member of a project team, managing the interface between computing, the organisation, and the end-users, recognising the different roles within a team and different ways of organising teams.	SC2	
LLO9	Select an appropriate range of tools and technologies and plan, organise, and implement a computing project involving fundamental game principles with due regard to human factors, user, and organisational requirements.	SC2	
LLO10	Carry out practical research, assisting organisations to undertake effective media application development.	SC2	
LLO11	Effectively communicate complex ideas and technical issues using verbal, visual and written techniques, and media appropriate to the target audience.	SC2	
LLO12	Analyse, evaluate, and interpret complex information and put such knowledge and understanding to practice in the design and implementation of computing applications, games, and media-based solutions for the individual users and/or organisational problems.	SC3	
LLO13	Recognise and analyse criteria and specifications appropriate to specific problems, and plan strategies for their solution.	SC3	
LLO14	Analyse and evaluate the extent to which a game-based artefact meets the criteria defined for its current use and predicted future requirements.	SC3	
LLO15	Select and deploy appropriate theory, tools and practice for the specification, design, implementation, and quality assurance of artefacts that utilise game development techniques.	SC3	
LLO16	Research and assess the use of new technologies and develop imaginative and practical game-based solutions to a diverse range of problems and challenges.	SC3	
	Research, gather and collate data from a range of sources, establish relevant connections, and apply to a given context to inform the		

LLO17	generation of design proposals.	SC3	
LLO18	Plan and manage work appropriately with an adherence to deadlines and project tracking.	SC3	
LLO19	Develop an understanding of personal learning needs and methods.	SC3	
LLO20	Use effective means of communication to express complex ideas about games design and development through appropriate means, based on considerations of context.	SC4	
LLO21	Make effective use of specialist ICT facilities concerned with the management of data and information, develop appropriate skills for this purpose, and adjust features to suit purpose.	SC4	
LLO22	Demonstrate knowledge and understanding through investigative problem-solving activities.	SC4	
LLO23	Present, formally, and informally, information about specialised games design and development topics to informed audiences.	SC4	
LLO24	Communicate with peers, senior colleagues, and specialists on a professional level.	SC4	
LLO25	Select, interpret, use, and evaluate a wide range of numerical and graphical data to set and achieve contextual goals, and to use this information as evidence when presenting arguments for decisions.	SC4	
LLO26	Adopt a broad-ranging and flexible approach to study by identifying learning needs and pursuing activities designed to meet these needs in increasingly autonomous ways.	SC5	
LLO27	Recognise the need for lifelong learning and undertake appropriate continued professional development, taking control of the learning process including organisational skills, time, and workload management.	SC5	
LLO28	Work effectively both independently and as a member of a team, showing a willingness to learn from others.	SC5	
LLO29	Maintain professional standards and quality control as applied to the specification, production, implementation and maintenance of computing and game-based applications.	SC5	
LLO30	Be innovative, foster creativity and drive, and manage the process of change.	SC5	
LLO31	Describe a basic knowledge and understanding of the basic context of games design and development, and its relationship to other subject areas such as design, technology, and engineering.	SC1	SCQF 7
LLO32	Outline familiarity with the basic facts and principles of computing and supporting topics via demonstrations of skills gained, the ability to answer questions and to describe and employ basic tools and techniques.	SC1	SCQF 7
LLO33	Express a working understanding of basic facts and principles in the field of games design and development through the construction of reports, solutions to standard problems, projects on the fundamentals of computing and creative design, and appropriate use of the conventions used in the topics outlined above.	SC1	SCQF 7
LLO34	Carry out study of the subject in a systematic, directed way with the aid of appropriate tutor guidance within the classroom and laboratory.	SC2	SCQF 7
LLO35	Apply the knowledge, skills and understanding of computing and games design and development subjects in practical contexts.	SC2	SCQF 7
LLO36	Employ some of the basic and routine professional skills, techniques, and practices associated with modern computing, with an appreciation for Human Computer Interaction and the extensive scope of this area.	SC2	SCQF 7
LLO37	Apply these skills in both routine and non-routine contexts.	SC2	SCQF 7
LLO38	Apply the basic principles and procedures found in computing and games design and development to standard, simple situations.	SC3	SCQF 7
LLO39	Present the ability to analyse situations and problems, based on the classification schemes presented by tutors.	SC3	SCQF 7

LLO40	Systematically relate to a limited number of facts/ideas/elements in an imitative manner.	SC3	SCQF 7
LLO41	Define their own value judgements based on criteria provided.	SC3	SCQF 7
LLO42	Convey aspects of computing concisely, in writing and orally, in relatively informal and limited-length pieces of work. Students should also be able to communicate appropriately the basic design and documentation of programs.	SC4	SCQF 7
LLO43	Achieve basic numeracy, mathematics, and the ability to manipulate data at a level appropriate to this stage of the degree. Students should also develop an understanding of basic concepts in logic and computation with an appreciation for databases and their wider use.	SC4	SCQF 7
LLO44	Obtain practical skills in computing using basic ICT applications, especially skills necessary for efficient use of games design and development.	SC4	SCQF 7
LLO45	Manage learning in an increasingly effective and purposeful way, showing the beginnings of development as an autonomous learner, and exercise some initiative and independence in carrying out defined activities at a professional level of practice.	SC5	SCQF 7
LLO46	Carry out interactions effectively with tutors and fellow students via software and web development tasks, problem-solving tutorials, and the production of appropriate documentation, and accept supervision in less familiar areas of work.	SC5	SCQF 7
LLO47	Exercise some managerial or supervisory responsibility for the work of others within a defined and supervised structure while managing limited resources within defined areas of work.	SC5	SCQF 7
LLO48	Take the lead in implementing agreed plans in familiar or defined contexts, while taking account of own and others' roles and responsibilities when carrying out and evaluating tasks.	SC5	SCQF 7
LLO49	Work, under guidance, with others to acquire an understanding of current professional practice.	SC5	SCQF 7
LLO50	Report an understanding for the subject core areas at this Stage of the course, specifically, games design and associated development with the exploration of experimental game mechanics, game design and gameplay, quality assurance for games and 3D reconstructive techniques.	SC1	SCQF 8
LLO51	Infer a growing appreciation for the field of study and wider scope, enabling students to map knowledge into a coherent picture of the discipline.	SC1	SCQF 8
LLO52	Compare the significance of subject core and peripheral areas of the subject, suggesting opportunities to explore and extend the boundaries of the discipline.	SC1	SCQF 8
LLO53	Practice knowledge, skills and understanding in using a range of professional skills, techniques, and practices associated with the topics studied, some of which are at an advanced or complex level.	SC2	SCQF 8
LLO54	Show the ability to carry out routine lines of enquiry, development or investigation into professional level problems and issues in relation to games design and development.	SC2	SCQF 8
LLO55	Adapt routine practices within accepted standards.	SC2	SCQF 8
LLO56	Use standard principles and procedures to somewhat more demanding situations.	SC3	SCQF 8
LLO57	Demonstrate increasing ability to undertake classifications and critically analyse relatively simple situations.	SC3	SCQF 8
LLO58	Demonstrate an appreciation of the need for relating and marshalling a range of facts/ideas/elements in an argued case.	SC3	SCQF 8
LLO59	Compare a range of valid alternative solutions, and begin to discriminate and evaluate in a reasoned, systematic, and increasingly independent way, both on an individual and group basis.	SC3	SCQF 8
LLO60	Use given tools/methods accurately and carefully to more demanding problems appropriate to this Stage.	SC3	SCQF 8
LLO61	Use communication skills in more formal Academic and professional styles, and in more extended pieces of work. Some presentation work may be introduced through the optional placement process.	SC4	SCQF 8

LLO62	Undertake the application of more advanced numerical, prototyping and design skills, relevant to specialised areas of games design and development, including; experimental game mechanics, game design and gameplay, quality assurance for games and 3D reconstructive techniques.	SC4	SCQF 8
LLO63	Use a broad selection of more advanced computing and IT skills in a wide range of contexts in all topics, but particularly in the development of solutions to Games Design problems and challenges.	SC4	SCQF 8
LLO64	Practice autonomy and initiative in some activities at a professional level within the range of exercises set by tutors.	SC5	SCQF 8
LLO65	Practice effective interactions within a learning or subject-specific group, demonstrating basic negotiating, role-ownership, leadership, debate /discussion, and group-support skills.	SC5	SCQF 8
LLO66	Plan work activities in familiar or defined contexts.	SC5	SCQF 8
LLO67	Practice management of resources and exercise responsibility for the work of others within a group and/or defined activity.	SC5	SCQF 8
LLO68	Plan ways that might show more awareness of own and others' roles, responsibilities and contributions when carrying out and evaluating tasks.	SC5	SCQF 8
LLO69	Work, under guidance, with others to acquire an understanding of current legal, ethical, and professional issues, and manage these issues in accordance with current professional and/or ethical codes of practices.	SC5	SCQF 8
LLO70	Review knowledge and understanding for the subject core areas at this Stage of the degree; gamification practices, 3D modelling, experiential development, and immersive technology.	SC1	SCQF 9
LLO71	Interpret a critical understanding of a range of the principles, main theories, concepts and terminology in the study and application of computing and games design and development techniques.	SC1	SCQF 9
LLO72	Estimate confident familiarity within the broad area of study and sufficient knowledge within the topics that explore various forms of media, including key specialist areas.	SC1	SCQF 9
LLO73	Estimate an understanding of limitations within such a broad area of study, together with an ability to relate the different areas to one another in a systematic and logical manner.	SC1	SCQF 9
LLO74	Demonstrate knowledge, skills and understanding in using a range of the principal professional skills, techniques, practices and/or materials associated with games design and development.	SC2	SCQF 9
LLO75	Demonstrate an applied knowledge of some specialised or advanced skills, techniques, practices and/or materials in the implementation of game artefacts.	SC2	SCQF 9
LLO76	Demonstrate application of routine methods of enquiry and research.	SC2	SCQF 9
LLO77	Demonstrate the ability to work in a range of professional contexts that include a degree of unpredictability.	SC2	SCQF 9
LLO78	Demonstrate the ability to select and use principles and procedures appropriate to the situation or problem in hand	SC3	SCQF 9
LLO79	Point out familiar ideas or principles in new contexts or situations and analyse systematically and effectively.	SC3	SCQF 9
LLO80	Assemble facts, ideas and elements in an argued case and relate game concepts and ideas to a wider range of situations.	SC3	SCQF 9
LLO81	Differentiate among a range of valid alternative solutions; begin to discriminate, and evaluate in a reasoned, systematic, and increasingly independent way using a range of sources.	SC3	SCQF 9
LLO82	Draw on established problem-solving skills to begin self-direct problem solving, including differentiating key elements of problems/challenges, and choose appropriate methods to construct their solution, with only general guidance in all topics.	SC3	SCQF 9
	Judge effectively in a variety of roles, to produce clear, well-structured reports and presentations on relevant aspects of games design and		

LLO83	development.	SC4	SCQF 9
LLO84	Interpret a selection of specialist numerical, modelling and design skills, relevant to the design and implementation procedures incurred in a game directed artefact, and the management of scope and progress.	SC4	SCQF 9
LLO85	Estimate and successfully use a broad selection of specialist computing and ICT skills in a wide range of contexts, in all topics.	SC4	SCQF 9
LLO86	Demonstrate a broad-ranging and flexible approach to study, identifying learning needs and pursuing activities designed to meet these needs in increasingly autonomous ways.	SC5	SCQF 9
LLO87	Seek an appropriate role in the management and development of a project, highly relevant or loosely in alignment to the subject of games design and development.	SC5	SCQF 9
LLO88	Assess, and be responsible for, the development and use of game applications.	SC5	SCQF 9
LLO89	Judge negotiation and collaboration skills when managing the interface between computing, games design and development, and other subject areas of an organisation, and the users of its computer systems.	SC5	SCQF 9
LLO90	Engage with legal, ethical, and professional issues in accordance with current codes of best practice, seeking guidance where appropriate.	SC5	SCQF 9
LLO91	Illustrate an understanding of the context of, and relationship between, the principal subject core areas, features, boundaries, terminology, and conventions of specialist areas of the course; AI for games, games development, and human computer interaction.	SC1	SCQF 10
LLO92	Illustrate confident familiarity and critical understanding of the defining concepts and features of Games Design, including further areas of special relevance or interest, and demonstrate a detailed understanding of the ways in which the subject is developed, including a range of established techniques of enquiry or research methodologies.	SC1	SCQF 10
LLO93	Develop appropriate high-level expertise in relevant design and implementation skills, and apply specialist analysis within a project setting, whilst learning technical skills appropriate to the specific task.	SC1	SCQF 10
LLO94	Communicate detailed knowledge and understanding in one or more specialisms related to the subject of games design and development, some of which is informed by, or at the forefront of the subject.	SC1	SCQF 10
LLO95	Conceptualise a clear understanding of topical boundaries, of the limits of application, and of the range of methods of study and types of judgement employed.	SC1	SCQF 10
LLO96	Execute knowledge, skills and understanding in using a wide range of the principal professional skills, techniques, and practices associated with games design and development.	SC2	SCQF 10
LLO97	Illustrate practical skills in the use of techniques and materials that are specialised, advanced and at the forefront of the subject of media-based systems.	SC2	SCQF 10
LLO98	Illustrate application of practical knowledge relevant to games design and development to execute a defined project of research, development, and investigation in the capstone project module.	SC2	SCQF 10
LLO99	Operate in a range of professional level contexts that include a greater degree of unpredictability within the overall games design and development specialism.	SC2	SCQF 10
LLO100	Compose an accurate selection and application of principles and procedures to the solution of a range of professional situations and problems, working autonomously, particularly within capstone project.	SC3	SCQF 10
LLO101	Examine principles and ideas in new contexts and situations, analysing them systematically, effectively, and critically, particularly within the capstone project.	SC3	SCQF 10

LLO102	Compose facts, ideas, and elements in support of a well-structured argument, design novel solutions and evolve new concepts, working autonomously, particularly within the capstone project.	SC3	SCQF 10
LLO103	Reconcile theory with professional and vocational practice, autonomously evaluate theory, process, solutions, and outcomes critically and effectively, particularly within the capstone project.	SC3	SCQF 10
LLO104	Develop creative solutions to complex problems and substantial challenges in an increasingly independent, confident, and flexible way and apply knowledge and skills appropriate to their solution, at the threshold of professional competence in all topics, but, particularly within the capstone project.	SC3	SCQF 10
LLO105	Justify the use of a broad selection of specialist computing and games design and development skills, in all topics, but particularly within the capstone project.	SC4	SCQF 10
LLO106	Communicate their effective use of specialist computing, design and ICT facilities concerned with the specialism of games, the visualisation /management of data and information, and develop appropriate skills for this purpose.	SC4	SCQF 10
LLO107	Compose effective means of communication to express complex ideas to colleagues through appropriate media, based on considerations of context.	SC4	SCQF 10
LLO108	Engage effectively in independent roles, debate in a professional manner, and produce detailed critiques and coherent project reports.	SC4	SCQF 10
LLO109	Communicate confidently with informed audiences in a range of contexts via oral and formal presentations, particularly within the capstone project.	SC4	SCQF 10
LLO110	Critique demonstrated skills, knowledge and understanding through autonomous investigative problem-solving activities, and manage the process relating to the development, maintenance, and review of games applications and contexts.	SC5	SCQF 10
LLO111	Argue the extent to which they maintain standards and manage quality control as applied to the specification, design, implementation, quality assurance and maintenance of game applications, in accordance with current and complex professional and ethical codes of practices and make informed judgments on issues not addressed by current professional and/or ethical codes or practices.	SC5	SCQF 10
LLO112	Critique a chosen approach to independent learning when provided with minimal guidance, using a wide range of resources appropriate to the profession, seeking, and making effective use of feedback, leading to personal development. At this level within all topics, but particularly within the capstone project.	SC5	SCQF 10
LLO113	Interact effectively within groups in all topics, and, where appropriate, demonstrate interaction and negotiating skills with users and other stakeholders.	SC5	SCQF 10
LLO114	Reconcile skills, knowledge and understanding through autonomous investigative problem-solving activities, and manage the process relating to the development, maintenance, and review of game specific artefacts and the wider subject of computing in this context.	SC5	SCQF 10

Programme Outcome Types Reference		
#	SCQF Characteristic	SCQF Level
SC1	Knowledge and understanding	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12

SC2	Practice: Applied knowledge, skills and understanding	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC3	Generic cognitive skills	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC4	Communication, numeracy and IT skills	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC5	Autonomy, accountability and working with others	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12

Delivery Range

<i>Title</i>	<i>Description</i>
Full Time On Campus	Placement and Study Abroad modules provide an additional option for students, which is not required to progress to the next stage and the credits accumulated do not contribute to the award total. CM1112 Introduction to Programming is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Stage 1 / Semester 1**Full Time On Campus**

Placement and Study Abroad modules provide an additional option for students, which is not required to progress to the next stage and the credits accumulated do not contribute to the award total. CM1112 Introduction to Programming is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CM1108	Creative Design		Jay Munro	Yes	2.0	15	Level 7
Full Time On Campus	CM1119	Problem Solving and Maths		Mark Snaith (ms10541)	Yes	2.0	15	Level 7
Full Time On Campus	CE1337	Programming Bootcamp		Mark Snaith (ms10541)	Yes	1.0	15	Level 7
Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	0739 - Electives 1	Games Design and Development - Electives Group 1		Jay Munro		1.0	15	

Stage 1 / Semester 2**Full Time On Campus**

Placement and Study Abroad modules provide an additional option for students, which is not required to progress to the next stage and the credits accumulated do not contribute to the award total. CM1112 Introduction to Programming is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>

Full Time On Campus	CM1120	Database Systems		Tiffany Young (ty7997)	Yes	3.0	15	Level 7
Full Time On Campus	CM1113	Software Design and Development		Mark Snaith (ms10541)	Yes	3.0	15	Level 7
Full Time On Campus	CM1121	Web UX		Jay Munro	Yes	3.0	15	Level 7
Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	0739 - Electives 2	Games Design and Development - Electives Group 2		Jay Munro		1.0	15	

Stage 2 / Semester 1**Full Time On Campus**

Placement and Study Abroad modules provide an additional option for students, which is not required to progress to the next stage and the credits accumulated do not contribute to the award total. CM1112 Introduction to Programming is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CM2119	Experimental Game Mechanics		Jay Munro	Yes	1.0	15	Level 8
Full Time On Campus	CM2120	Game Design and Gameplay		Jay Munro	Yes	1.0	15	Level 8
Full Time On Campus	CM2112	Real World Projects and Professional Skills		Ross McLean (rjm7342)	Yes	1.0	15	Level 8
Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	0739 - Electives 3	Games Design and Development - Electives Group 3		Jay Munro		1.0	15	

Stage 2 / Semester 2**Full Time On Campus**

Placement and Study Abroad modules provide an additional option for students, which is not required to progress to the next stage and the credits accumulated do not contribute to the award total. CM1112 Introduction to Programming is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CM2121	3D Reconstructive Techniques		Jay Munro	Yes	1.0	15	Level 8
Full								

Time On Campus	CM2122	Quality Assurance for Games		Jay Munro	Yes	1.0	15	Level 8
Full Time On Campus	CM2124	Research Methods		Daniel Gardner	Yes	2.0	15	Level 8
Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	0739 - Electives 4	Games Design and Development - Electives Group 4		Jay Munro		1.0	15	

Stage 2 / Semester 3**Full Time On Campus**

Placement and Study Abroad modules provide an additional option for students, which is not required to progress to the next stage and the credits accumulated do not contribute to the award total. CM1112 Introduction to Programming is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	0739 - Electives 5	Games Design and Development - Electives Group 5		Jay Munro		1.0	90	

Stage 3 / Semester 1**Full Time On Campus**

Placement and Study Abroad modules provide an additional option for students, which is not required to progress to the next stage and the credits accumulated do not contribute to the award total. CM1112 Introduction to Programming is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CM3112	3D Modelling			Yes	1.0	15	Level 9
Full Time On Campus	CM3137	Gamification Practices		Jay Munro	Yes	1.0	15	Level 9
Full Time On Campus	CM3130	Interdisciplinary Team Project		Mark Zarb (mz7371)	Yes	1.0	15	Level 9

Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	0739 - Electives 6	Games Design and Development - Electives Group 6		Jay Munro		1.0	15	

Stage 3 / Semester 2**Full Time On Campus**

Placement and Study Abroad modules provide an additional option for students, which is not required to progress to the next stage and the credits accumulated do not contribute to the award total. CM1112 Introduction to Programming is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CE3025	Experiential Development		Jay Munro	Yes	1.0	15	Level 9
Full Time On Campus	CM3150	Immersive Technology		Jay Munro	Yes	2.0	15	Level 9
Full Time On Campus	CM3141	Pirate Studies		Ross McLean (rjm7342)	Yes	1.0	15	Level 9
Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	0739 - Electives 7	Games Design and Development - Electives Group 7		Jay Munro		1.0	15	

Stage 3 / Semester 3**Full Time On Campus**

Placement and Study Abroad modules provide an additional option for students, which is not required to progress to the next stage and the credits accumulated do not contribute to the award total. CM1112 Introduction to Programming is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	0739 - Electives 5	Games Design and Development - Electives Group 5		Jay Munro		1.0	90	

Stage 4 / Semester 1**Full Time On Campus**

Placement and Study Abroad modules provide an additional option for students, which is not required to progress to the next stage and the credits accumulated do not contribute to the award total. CM1112 Introduction to Programming is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CM4137	AI for Games		Jay Munro	Yes	1.0	15	Level 10
Full Time On Campus	CM4114	Games Development		Mark Zarb (mz7371)	Yes	2.0	15	Level 10
Full Time On Campus	CM4110	Human Computer Interaction			Yes	1.0	15	Level 10
Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	0739 - Electives 8	Games Design and Development - Electives Group 8		Jay Munro		1.0	15	

Stage 4 / Semester 2**Full Time On Campus**

Placement and Study Abroad modules provide an additional option for students, which is not required to progress to the next stage and the credits accumulated do not contribute to the award total. CM1112 Introduction to Programming is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CE4321	Capstone Project		Mark Zarb (mz7371)	Yes	1.0	60	Level 10

Course Deliveries

Stage	Semester	Delivery Type	Delivery Range	Code	Title	Notes	Owner	Allow Mapping	Version	Credits	Module Level
1	Semester 1	Core	Full Time On Campus	CM1108	Creative Design		Jay Munro	Yes	2.0	15	Level 7
1	Semester 1	Core	Full Time On Campus	CM1119	Problem Solving and Maths		Mark Snaith (ms10541)	Yes	2.0	15	Level 7
1	Semester 1	Core	Full Time On Campus	CE1337	Programming Bootcamp		Mark Snaith (ms10541)	Yes	1.0	15	Level 7
1	Semester 1	Elective	Full Time On Campus	0739 - Electives 1	Games Design and Development - Electives Group 1		Jay Munro		1.0	15	
1	Semester 2	Core	Full Time On Campus	CM1120	Database Systems		Tiffany Young (ty7997)	Yes	3.0	15	Level 7
1	Semester 2	Core	Full Time On Campus	CM1113	Software Design and Development		Mark Snaith (ms10541)	Yes	3.0	15	Level 7
1	Semester 2	Core	Full Time On Campus	CM1121	Web UX		Jay Munro	Yes	3.0	15	Level 7
1	Semester 2	Elective	Full Time On Campus	0739 - Electives 2	Games Design and Development - Electives Group 2		Jay Munro		1.0	15	
2	Semester 1	Core	Full Time On	CM2119	Experimental Game Mechanics		Jay Munro	Yes	1.0	15	Level 8

			Campus									
2	Semester 1	Core	Full Time On Campus	CM2120	Game Design and Gameplay		Jay Munro	Yes	1.0	15	Level 8	
2	Semester 1	Core	Full Time On Campus	CM2112	Real World Projects and Professional Skills		Ross McLean (rjm7342)	Yes	1.0	15	Level 8	
2	Semester 1	Elective	Full Time On Campus	0739 - Electives 3	Games Design and Development - Electives Group 3		Jay Munro		1.0	15		
2	Semester 2	Core	Full Time On Campus	CM2121	3D Reconstructive Techniques		Jay Munro	Yes	1.0	15	Level 8	
2	Semester 2	Core	Full Time On Campus	CM2122	Quality Assurance for Games		Jay Munro	Yes	1.0	15	Level 8	
2	Semester 2	Core	Full Time On Campus	CM2124	Research Methods		Daniel Gardner	Yes	2.0	15	Level 8	
2	Semester 2	Elective	Full Time On Campus	0739 - Electives 4	Games Design and Development - Electives Group 4		Jay Munro		1.0	15		
2	Semester 3	Elective	Full Time On Campus	0739 - Electives 5	Games Design and Development - Electives Group 5		Jay Munro		1.0	90		
3	Semester 1	Core	Full Time On Campus	CM3112	3D Modelling			Yes	1.0	15	Level 9	
			Full									

3	Semester 1	Core	Time On Campus	CM3137	Gamification Practices		Jay Munro	Yes	1.0	15	Level 9
3	Semester 1	Core	Full Time On Campus	CM3130	Interdisciplinary Team Project		Mark Zarb (mz7371)	Yes	1.0	15	Level 9
3	Semester 1	Elective	Full Time On Campus	0739 - Electives 6	Games Design and Development - Electives Group 6		Jay Munro		1.0	15	
3	Semester 2	Core	Full Time On Campus	CE3025	Experiential Development		Jay Munro	Yes	1.0	15	Level 9
3	Semester 2	Core	Full Time On Campus	CM3150	Immersive Technology		Jay Munro	Yes	2.0	15	Level 9
3	Semester 2	Core	Full Time On Campus	CM3141	Pirate Studies		Ross McLean (rjm7342)	Yes	1.0	15	Level 9
3	Semester 2	Elective	Full Time On Campus	0739 - Electives 7	Games Design and Development - Electives Group 7		Jay Munro		1.0	15	
3	Semester 3	Elective	Full Time On Campus	0739 - Electives 5	Games Design and Development - Electives Group 5		Jay Munro		1.0	90	
4	Semester 1	Core	Full Time On Campus	CM4137	AI for Games		Jay Munro	Yes	1.0	15	Level 10
4	Semester 1	Core	Full Time On Campus	CM4114	Games Development		Mark Zarb (mz7371)	Yes	2.0	15	Level 10

			Campus								
4	Semester 1	Core	Full Time On Campus	CM4110	Human Computer Interaction			Yes	1.0	15	Level 10
4	Semester 1	Elective	Full Time On Campus	0739 - Electives 8	Games Design and Development - Electives Group 8		Jay Munro		1.0	15	
4	Semester 2	Core	Full Time On Campus	CE4321	Capstone Project		Mark Zarb (mz7371)	Yes	1.0	60	Level 10

Delivery

Location	Mode of Study	Mode of Attendance	Teaching Period Key Date
Aberdeen, United Kingdom	On-Campus	Full Time	

Sources of Further Information

#	URL and Description
1	https://www.bcs.org/media/9006/course-accreditation-guidelines.pdf A number of external sources were consulted in the process of curriculum development for the BSc (Hons) Games Design and Development course. The accreditation guidelines for the BCS, The Chartered Institute for IT, Guidelines on course accreditation, Information for universities and colleges, January 2020, can be found at: https://www.bcs.org/media/9006/course-accreditation-guidelines.pdf and Academic Accreditation Guidelines, April 2022 can be found at: https://www.bcs.org/media/1209/accreditation-guidelines.pdf The Association for Computing Machinery (ACM) Guidelines on curricular elements for Computer Science (and related areas), Computer Science Curricula 2020, can be found at: https://www.acm.org/binaries/content/assets/education/curricula-recommendations/cc2020.pdf Further information about the Undergraduate Programme in the School of Computing can be found on the RGU website: https://www.rgu.ac.uk/study/types-of-study/undergraduate-courses/computing-undergraduate